

Pigeonhole & Extremal Reasoning

The common idea is structural inevitability. Instead of enumerating every configuration, compress the problem until failure would have too little capacity or would create an object more extreme than one already chosen.

1. Reconnect: capacity before formula

Put 13 objects into 12 boxes. If every box held at most one object, total capacity would be 12, not 13. Therefore a collision is unavoidable.

The reusable model is:

OBJECTS $\rightarrow$ BOXES $\rightarrow$ FAILURE CAPACITY $\rightarrow$ CONTRADICTION $\rightarrow$ TARGET RELATION

For example, among 21 integers there are two with the same remainder modulo 20: the 20 remainder classes are the boxes. Equal remainder then implies that their difference is divisible by 20.

2. Generalized pigeonhole

To force at least k objects into one of m boxes, negate the claim. If every box had at most $k-1$, the total could be at most $(k-1)m$. Any larger total forces a box with at least k objects.

This is stronger than an informal average slogan. The average may suggest the threshold; the proof is the integer capacity contradiction.

3. Box design is the creative step

The conclusion should suggest the boxes.

To prove that 11 chosen numbers from $\{1, \dots, 20\}$ include two consecutive numbers, use the ten boxes $\{1,2\}, \{3,4\}, \dots, \{19,20\}$. Eleven choices force two into one pair.

To prove a divisibility relation among selected numbers, residue classes or odd-part classes may be useful boxes. Arithmetic classification must already be justified; COMB-05 owns the collision step, not a new modular-arithmetic chapter.

4. Geometric pigeonhole

In geometry, a box is usually a cell in a partition. The crucial check is its diameter, not merely its area.

Five points in a unit square: divide the square into four half-size squares. Two points occupy one small square, whose diameter is $\sqrt{2}/2$; hence their distance is at most $\sqrt{2}/2$.

The design rule is:

1. decide the metric relation to force;
2. partition into cells where any same-cell pair automatically satisfies it;
3. compare the number of points with the number of cells;
4. assign boundary points consistently.

5. Extremal choice

Sometimes boxes are unnatural. In a finite configuration choose a smallest, largest, nearest, or farthest admissible object and exploit what its extremality forbids.

Suppose a finite set of real numbers is closed under taking the midpoint of any two distinct members. If there were two distinct members, choose the closest pair $x < y$. Their midpoint lies in the set and is closer to each endpoint than $y - x$, contradicting the choice. Thus the set has only one member.

The method is:

CHOOSE EXTREME OBJECT $\rightarrow$ PRESERVE CONSTRAINTS $\rightarrow$ PRODUCE STRICT IMPROVEMENT $\rightarrow$ CONTRADICTION

Naming the smallest object is not enough; the proof must use the no-improvement property.

6. Mandatory contrasts

Pigeonhole vs inclusion-exclusion

Pigeonhole answers an existence question: must some collision occur? Inclusion-exclusion answers an exact counting question with overlaps: how many outcomes satisfy at least one property? Do not use overlap machinery when capacity already proves existence, and do not expect pigeonhole to compute an exact union size.

Extremal choice vs inequality optimization

Extremal choice selects an object from a finite configuration and uses the fact that no admissible object can be more extreme. Inequality optimization bounds a numerical expression over a continuous or algebraic domain. The words maximum/minimum alone do not decide the method.

Counting average vs structural inevitability

An average is a numerical summary. Pigeonhole uses the stronger discrete statement that if every box stayed below a threshold, the total capacity would be insufficient.

7. Historical anchor: extremal complement

For IOQM-2023-Q27, the universe contains 4845 increasing quadruples. Exactly 525 are balanced, leaving 4320 unbalanced. A chosen family of 4411 therefore forces at least $4411 - 4320 = 91$ balanced members. The first move is to maximize the bad class globally, not inspect the chosen family one member at a time.

8. Historical anchor: local crossing cap to global density

For IOQM-2023-Q18, a local condition says each selected diagonal of a convex 50-gon crosses at most one other selected diagonal. The successful representation is a sparse crossing graph/drawing. Teacher-side source analysis gives the tight upper bound 71 and a matching construction. The learner lesson is to translate a local restriction into a global extremal structure before greedy addition.

9. Adopt the router

When a problem asks what must happen, ask:

- What are the objects?
- What boxes make the desired relation automatic after a collision?
- What capacity would failure impose?
- If boxes do not simplify the problem, which extreme object has a useful no-improvement consequence?
- Is the task actually an exact count or a continuous inequality optimum instead?

Write the structural first line before doing arithmetic.

COMB-05 - First-Step Reference

One router

FORCED EXISTENCE? -> CHOOSE BOXES -> COMPARE OBJECTS WITH CAPACITY -> FORCE COLLISION

If the problem asks for a largest/smallest possible structure rather than mere existence, switch to:

CHOOSE AN EXTREME OBJECT -> ASK WHAT ITS EXTREMENESS FORBIDS -> COUNT/CONTRADICT -> CHECK SHARPNESS

Recognition atlas

Surface clue	Structural question	First useful line
more objects than categories	what are the categories?	define objects and boxes explicitly
at least $k+1$ in one class	what is the maximum allowed per box if the claim fails?	failed claim gives at most k per box
distances / intervals / cells	can the region be partitioned into small-diameter boxes?	state the partition and its diameter bound
residues / divisibility	which residue or divisor-signature classes make equal-box membership useful?	
name the residue classes		
largest / smallest / nearest / farthest	what local consequence follows from that choice?	choose the extreme object and write the forbidden improvement
exact count requested	is existence enough?	if not, retrieve direct counting/complement/IE instead

Three mandatory boundaries

Pigeonhole vs inclusion-exclusion. Pigeonhole proves that some collision must occur; inclusion-exclusion computes the size of a union or overlap-corrected count. If no exact number of outcomes is requested, try inevitability first.

Extremal choice vs inequality optimization. Extremal reasoning chooses an object from a finite configuration and exploits what would improve it. Inequality optimization bounds a numerical expression. Do not replace a structural contradiction with AM-GM merely because the word maximum appears.

Counting average vs structural inevitability. An average can suggest a threshold, but the proof is the capacity statement: if every box stayed below the threshold, the total capacity would be too small.

Quick checks

- Are the boxes exhaustive?
- Are they disjoint, or have you stated how ties are assigned?
- Is the capacity bound correct?
- Does the collision actually imply the target relation?
- In an extremal proof, did you use the extremal property, not merely name an extreme object?

COMB-05 - Recognition and First-Line Lab

Attempt each item before reading any later solution. For items 1-8, write only the first useful mathematical line or structural setup.

1. Seventeen integers are chosen. Show that two have the same remainder modulo 16.

2. Thirty-one students choose one of 10 clubs. Show that some club has at least 4 of these students.
3. Nine points lie in a unit square. Give a partition that forces two points to be at distance at most $\sqrt{2}/2$.
4. Six integers are chosen from $\{1,2,\dots,10\}$. Choose boxes that force two whose sum is 11.
5. A finite set of positive integers has the property that no member divides another. Which kind of extremal element is useful if one tries to enlarge the set under an extra divisibility closure condition? State the choice, not a full proof.
6. A collection of intervals on a line is known to have pairwise-overlap restrictions. The target is a contradiction from the shortest interval. What is the first object you select?
7. A problem asks for the number of 6-digit strings avoiding three forbidden properties. Should your first reflex be pigeonhole or inclusion-exclusion? State the structural reason.
8. A problem asks for the minimum possible value of x^2+y^2 under a real constraint. Should your first reflex be extremal-choice reasoning or inequality optimization? State the structural reason.

Full short solves

9. Prove that among any 13 integers there are two whose difference is divisible by 12.
10. Place 10 points in an equilateral triangle of side 1. Prove that two are at distance at most $1/3$.
11. Choose 51 integers from $\{1,2,\dots,100\}$. Prove that two chosen integers are consecutive.
12. From any 6 people, prove that there are either 3 mutual acquaintances or 3 mutual strangers, assuming acquaintance is symmetric.

COMB-05 - Practice and Transfer Bank

F0 - reconnect

1. Put 25 balls into 6 boxes. Prove that one box contains at least 5 balls.
2. Among 8 integers, prove that two have the same remainder modulo 7.

F1 - direct and generalized pigeonhole

3. Among 101 integers, prove that two have a difference divisible by 100.
4. Forty students are assigned to 12 months by birth month. Prove that some month has at least 4 birthdays.
5. Choose 11 integers from $\{1,\dots,20\}$. Prove that two are consecutive.

F2 - geometric and number-theoretic surfaces

6. Five points lie in a unit square. Prove that two are at distance at most $\sqrt{2}/2$.
7. Ten points lie in an equilateral triangle of side 1. Prove that two are at distance at most $1/3$.
8. Choose 7 integers from $\{1,\dots,12\}$. Prove that two have sum 13.
9. Among any 5 integers, prove that two have a difference divisible by 4.

F3 - extremal choice

10. In any finite set of at least two distinct real numbers, choose the closest pair. Explain why no third selected point can lie strictly between them at smaller distance from one endpoint.
11. A finite graph has at least one edge. Choose a vertex of largest degree. State the immediate degree comparison available against every other vertex.
12. A finite family of nonempty sets is partially ordered by inclusion. Choose a set of smallest cardinality. Explain what would contradict this choice.

F4 - changed-surface transfer

13. Fifty-one distinct integers are selected from $\{1,\dots,100\}$. Prove that two are consecutive. Then explain why exact enumeration of all 51-subsets is irrelevant.
14. Twenty-five points lie in a square of side 2. Prove that two are at distance at most $\sqrt{2}/2$.
15. A party has 6 people. Prove that there are three mutual acquaintances or three mutual strangers.
16. A collection of 4411 increasing quadruples (a,b,d,c) is chosen from $1 \leq a < b < d < c \leq 20$. Call a quadruple balanced when $a+c=b+d$. Prove that at least 91 chosen quadruples are balanced.

Why-not / verification

17. Why does an average of 3.2 objects per box not by itself prove a particular box has exactly 4 objects?
18. Why is inclusion-exclusion oversized for: 'among 13 integers, two differ by a multiple of 12'?
19. Why is AM-GM not the canonical first move in a finite configuration problem that says 'choose the shortest edge and derive a crossing contradiction'?
20. In a geometric pigeonhole proof, why must the diameter of each cell be checked rather than only its area?

COMB-05 - Mixed Mastery Test

No method labels or default hints are supplied on the first attempt.

1. Twenty integers are written on a board. Prove that two have the same remainder when divided by 19.
2. Thirty-seven objects are distributed among 9 containers. Prove that one container receives at least 5 objects.
3. Choose 8 integers from $\{1,2,\dots,14\}$. Prove that two sum to 15.
4. Five points lie in a unit square. Prove that two are at distance at most $\sqrt{2}/2$.
5. Choose 51 distinct integers from $\{1,\dots,100\}$. Prove that two are consecutive.
6. Among any 6 people, prove that there are three mutual acquaintances or three mutual strangers.
7. A finite nonempty set of positive real numbers is closed under replacing two unequal members by their geometric mean. Explain why choosing a smallest member is not, by itself, a useful extremal step; identify the missing contradiction mechanism.
8. A problem asks for the exact number of arrangements satisfying at least one of three overlapping

restrictions. Explain why a pigeonhole proof cannot replace overlap-corrected counting.

9. A problem asks for a lower bound on a real expression subject to $xy=1$. Explain why selecting the smallest of finitely many objects is not the right model.

10. Historical-anchor audit: among all increasing quadruples from $\{1, \dots, 20\}$, balanced means $a+c=b+d$. Determine the number of balanced quadruples.

11. Historical-anchor consequence: if 4411 such quadruples are selected, what minimum number must be balanced?

12. Transfer: 17 points lie in a square of side 2. Prove that two points are at distance at most $\sqrt{2}$.